

Introduction to Data Science

Course 1: Welcome to Data Science and ML (IFT6758)

Glen Berseth

Université de Montréal and Mila Québec AI Institute

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Université
de Montréal



Mila

CIFAR

Outline

What is Data Science?

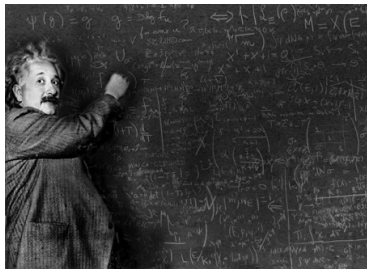
Data Science Requires Diverse Skills

Course Details

LLMs are Tools, Experts can Do Better With Tools

Data Science is not pure Machine Learning (IFT6390)

- Pure ML is like Pure Physics
- Asking: What is the grand unified theory of the universe?
 - Does the algorithm converge to a good fixed point?



- Data Science is like engineering
- Asking: How can we build a robot that looks like Einstein robot
 - How to construct a system to repeat this success reliably?

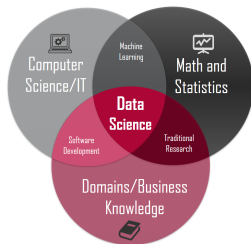


A possible Definition of Data Science

- “Data science is the application of computational and statistical techniques to address or gain insight into some problem in the real world”
- Source: Zico Kolter’s course on Data Science

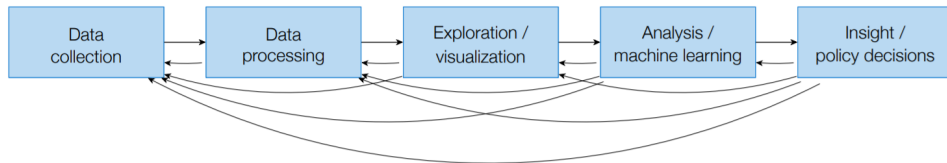
How do we get insight from data?

- Start with the Data
- What data is available, need more, pre-processing data,
- Analyze the data
- Which algorithm/model, how do I know it is working, how to improve it
- Scale the System (MLOps)
 - How to use compute, distribute experiments, version control (git), avoid errors



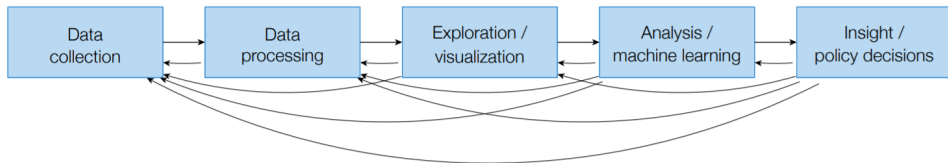
Example: Voice-to-text problem

- Why Voice to text?
 - To make a good model that you can use for yourself and others we need to have and use all the skills needed for data science



Example: Voice-to-text problem

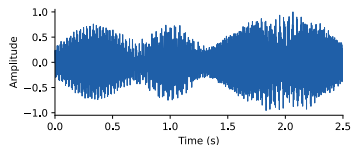
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- And why each step

Voice-to-text problem

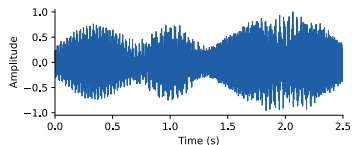
- What did the speaker say in this audio clip?
- Data:
 - Audio waveform
 - Speaker/accent metadata
- Analysis:
 - What should we predict?
 - How can we train a model to predict it?
 - How do we know the model is working?



- Transcript (y): “Bonjour, comment allez-vous?”

Voice-to-text problem

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- Analysis:
 - What should we predict?
 - How can we train a model to predict it?
 - How do we know the model is working?
- Scale:
 - We have thousands of hours of audio to transcribe
 - What compute should we use?
 - How can we make it easy to run this code in the cloud?

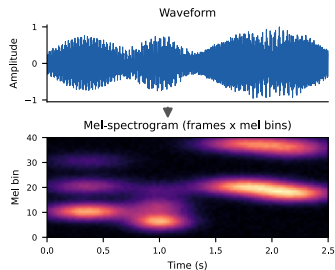


- Transcript (y): “Bonjour, comment allez-vous?”



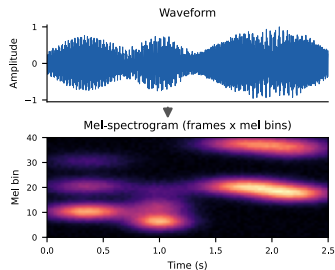
Data: What is available, what should be used?

- Audio waveform (x)
 - Raw sequence of audio samples, e.g. 16kHz
 - $[0.0021, -0.0034, 0.0012], \dots$
 - Converted to a mel-spectrogram (frames x mel bins)
 - $[f_1, f_2, f_3], \dots$
 - Turns audio into an image, reusing standard image architectures
 - Is this enough?



Data: What is available, what should be used?

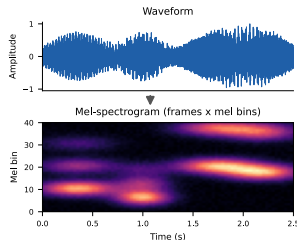
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 - [f_1, f_2, f_3], ...
 - Turns audio into an image, reusing standard image architectures
 - Is this enough?
- Feedback (y)
 - Ground-truth transcript
- Is this enough?
 - Want to predict the exact words spoken.
 - Add speaker accent, language, punctuation, etc.



- Transcript (y): “Bonjour, comment allez-vous?”

Analysis: Train a Model, What type of Prediction?

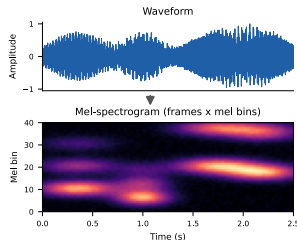
- What type is the input? (x)
 - Sequence of mel-spectrogram frames
 - $[f_1, f_2, f_3], \dots$
 - Plus speaker/accent metadata at times
 - $[\text{accent}, \text{fr-CA}], \dots$
 - Output y is a sequence of text tokens.
- Feedback (y)
 - Ground-truth transcript
 - Sequence-to-sequence prediction



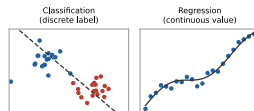
speaker metadata
speaker_id: spk_0342
accent: fr-CA
age: 34
gender: F

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- Possible Methods
 - Classification, Regression
 - Sequence-to-sequence (encoder-decoder)
 - Which one?

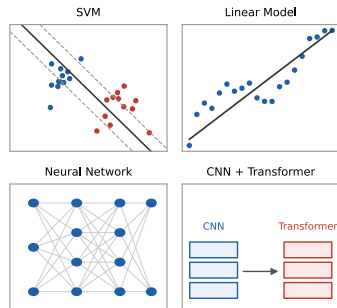


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Analysis: Which Classification Algorithm Works Best?

- What type of Model/Architecture?
- Support Vector machine
 - Theoretically sound
 - Does not scale well to large input sizes
- Linear Model
- Is a transcript a linear function of the raw waveform?
- No
- Neural Network
 - Sometimes unstable to train
 - Takes long time to train
- CNN + Transformer encoder-decoder (Whisper)
 - Scales well, this is what we will fine-tune



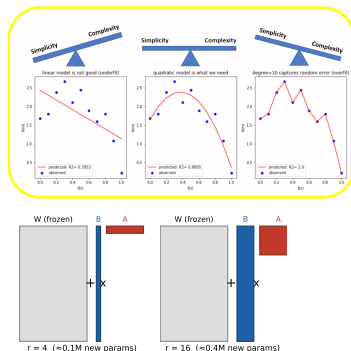
Underfitting/Overfitting: Is the Model fitting the data well?

- Underfitting
 - Linear models have little representation
- Overfitting
 - Model has learned special cases
 - These special cases are not part of real distribution
- How to know the fit is appropriate
 - Leave out some recorded audio for testing (validation split)
 - Cross Validation
- How to get a better fit
 - Regularization (e.g. LoRA dropout)
 - Occam's razor: simplest model is the best



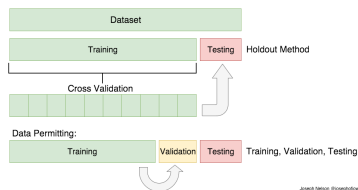
Model Complexity: And how to Measure it

- Decision Trees
 - Simple trees
- Deep networks
 - Number of parameters
 - Number of layers
- LoRA adapters (Hu *et al.*, 2022)
 - Rank r controls how many new parameters are added
 - $r=16$ keeps the base model mostly frozen



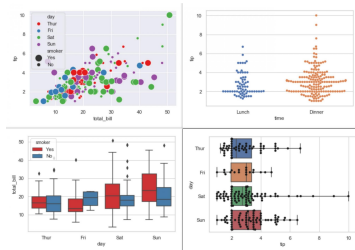
Pre-Processing and Model Tuning

- Is the model performing well enough?
 - We can often get better performance.
- Pre-process the data
 - How to split data for test/validation
 - Do we have enough data to split?
- Model Tuning:
 - All models have hyper parameters
 - Learning rate, batch size, LoRA rank/alpha
 - Try different hyperparameters values



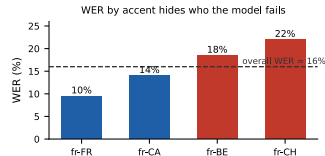
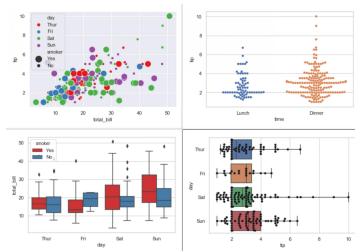
Data Visualization: Best method to display results

- How to show ourselves and others the model is working
- Exploration:
 - Are there other trends in the data?
 - How to find them?
- Presentation:
 - Best way to display the exploration?
 - Want a compact visualization with little noise



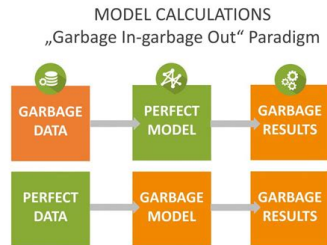
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- Advanced visualizations:
 - Multi-Modal visualization, e.g. spectrograms next to attention maps
 - Interactive visualizations
 - Traps of Visualization



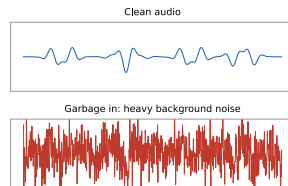
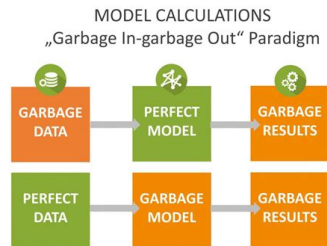
Data Preprocessing and Augmentation

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 - Garbage in garbage out
 - Filter out very long/short audio clips
 - Resample all audio to the model's expected rate (16kHz)



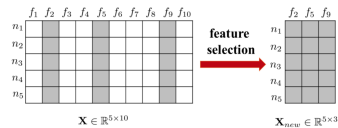
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- Augmentation:
 - Add background noise, speed/pitch perturbation
 - Others methos????



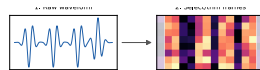
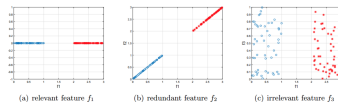
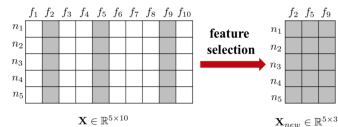
Feature Engineering: Selecting better features

- Not all features are Equal, use the better ones
 - Recall the transcription problem

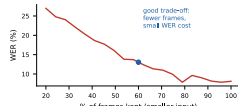


Feature Engineering: Selecting better features

- Not all features are Equal, use the better ones
 - Recall the transcription problem
- What is a feature:
 - X = mel-spectrogram frames $[f_1, f_2, f_3]$
 - Maybe the last few seconds of audio work best
- Selection:
 - Reduce input length
 - Possibly use a smaller/distilled model

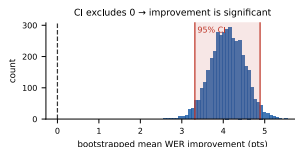
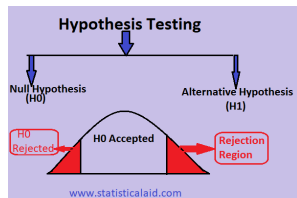


3. Effect on model performance



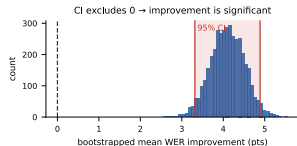
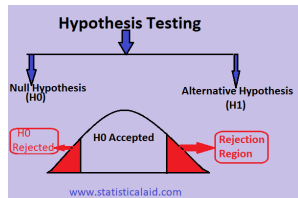
Hypothesis testing: Be confident in your model

- How confident are we in our results?
- Hypothesis Testing
 - Statistical test for likelihood of true
 - Vs null hypothesis
- Methods
 - Direct simulation
 - Shuffling
 - Bootstrapping
 - Cross Validation



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- Example: is fine-tuned Whisper actually better?
 - Compare baseline vs. fine-tuned WER, same test clips
 - Bootstrap the per-clip WER differences
 - CI excludes 0 $\Rightarrow$ real improvement



MLOps: Experiment Tracking and Production-Level Tools

- How many different models will we try and train? 100? 10,000?
 - A good scientist keeps a journal of all experiments
 - Note WHY each experiment was performed, etc.
 - After two weeks people all details on prior work.



Unsupervised Learning

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- Unsupervised learning looks for structure in the data itself, without any ground-truth targets to predict
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- Example: Whisper was pretrained this way before we fine-tuned it
 - Pretrained on 680,000 hours of weakly-labeled audio-text pairs
 - “Weakly-labeled” means the transcripts were not carefully verified by humans, they were scraped from the internet (e.g. video captions and subtitles), so they are noisy but plentiful
 - This gives the model broad speech patterns before we fine-tune it on a small, cleanly-labeled dataset

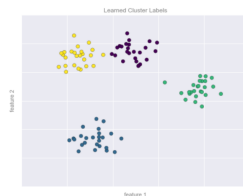
Unsupervised Learning: Clustering

- What can we do when we don't have the labels y ?
 - Hypothesis: points with the same label should be similar
 - Clustering: a model that aims at regrouping “similar” input data points
 - The complexity boils down to how we define “similar” (which distance? which features?)



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 - The complexity boils down to how we define “similar” (which distance? which features?)
- Example: cleaning the Whisper fine-tuning dataset
 - Cluster audio clips using extracted features (e.g. embeddings, SNR, duration)
 - Small/outlier clusters often correspond to corrupted audio or mislabeled transcripts
 - Lets us flag and remove bad clips before fine-tuning, without needing new labels



Algorithmic Bias

- How do we know the predictions are fair?
 - Really is the data fair?
 - Does the data represent the true causal relation:
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- Fairness:
 - Understand the effects of these issues
 - Make you model and data more fair
 - Avoid amplifying historical bias



Hardware and Distributed learning

- What hardware should we buy to run our algorithm?
 - Need to justify this to the boss
- Hardware:
 - GPU needed for audio processing
 - How many queries / minute?
 - LoRA + mixed precision (fp16) lets us fine-tune on a single GPU
 - GPU hardware precision leads to **different results**. (fp32 or fp16 or pf8) (Li *et al.*, 2024; Khatri *et al.*, 2026)
- Distribute:
 - Which cloud computer to use?
 - Depends on model assumptions

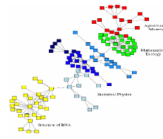


Graph Data

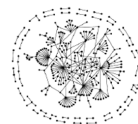
- Specificity of Graphs
 - Standard tasks
 - representations
 - Adjacency matrix, Laplacian matrix
- Node embeddings
 - By node similarity
 - Random walks



Social networks



Economic networks



Biomedical networks



Information networks:
Web & citations



Internet



Networks of neurons

Graph Data

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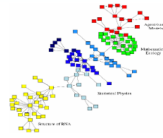
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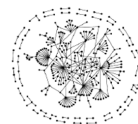
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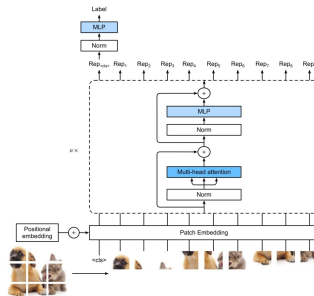


Networks of neurons

- Example: Social network

- Node information: location, age, etc.
- Link information: friends, likes

- Specificity of Graphs
 - How to create an LLM [Li *et al.* \(2024\)](#); [Khatri *et al.* \(2026\)](#)
 - Tokenization (feature engineering)
 - Supervised fine tuning (SFT)
- Reinforcement Learning
 - Reinforcement learning from human feedback
 - Using verified rewards
- Vision Language Models
 - Tokenizing multiple data types
 - Training



Digging deeper

- By the end of this class you will understand the why and how to build such systems in a solid and professional way.

Data Science Requires Diverse Skills

MODERN DATA SCIENTIST

Data Scientist, the sexiest job of 21st century requires a mixture of multidisciplinary skills ranging from an intersection of mathematics, statistics, computer science, communication and business. Finding a data scientist is hard. Finding people who understand who a data scientist is, is equally hard. So here is a little cheat sheet on who the modern data scientist really is.

MATH & STATISTICS

- ☆ Machine learning
- ☆ Statistical modeling
- ☆ Experiment design
- ☆ Bayesian inference
- ☆ Supervised learning: decision trees, random forests, logistic regression
- ☆ Unsupervised learning: clustering, dimensionality reduction
- ☆ Optimization: gradient descent and variants



The illustration shows a man and a woman standing side-by-side. The man is wearing a dark blue suit, a white shirt, and a red tie. The woman is wearing a red top and a dark blue skirt. Both are wearing glasses and have a professional appearance.

PROGRAMMING & DATABASE

- ☆ Computer science fundamentals
- ☆ Scripting language e.g. Python
- ☆ Statistical computing package e.g. R
- ☆ Database: SQL and NoSQL
- ☆ Relational algebra
- ☆ Parallel databases and parallel query processing
- ☆ MapReduce concepts
- ☆ Hadoop and HIVE/Hig
- ☆ Custom reducers
- ☆ Experience with xaaS like AWS

DOMAIN KNOWLEDGE & SOFT SKILLS

- ☆ Passionate about the business
- ☆ Cares about data
- ☆ Influence without authority
- ☆ Hacker mindset
- ☆ Problem solver
- ☆ Strategic, proactive, creative, innovative and collaborative

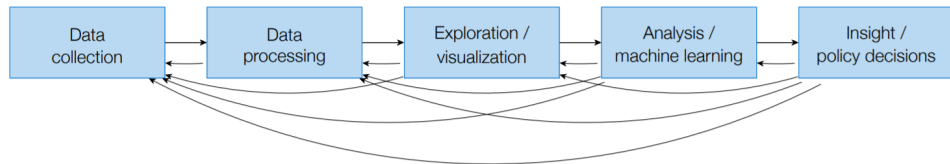
COMMUNICATION & VISUALIZATION

- ☆ Able to engage with senior management
- ☆ Story telling skills
- ☆ Translate data driven insights into decisions and actions
- ☆ Visual art design
- ☆ R packages like ggplot or lattice
- ☆ Knowledge of any of visualization tools e.g. Flare, ECLIP, Tableau

- Algorithmic Justice League



Back to what is Data science



- Source: <https://www.datasciencecourse.org/slides/intro.pdf>

First Step: Webpage of the course

- Static page of the course:
 - French section: <https://fracturedplane.notion.site/Cours-de-Science-des-Donn-es-IFT6758B-2026-3c1214857276818499f7e8eee49dc68a?pvs=74>
 - English section: <https://fracturedplane.notion.site/Data-Science-Course-IFT6758B-2026-3ae21485727681a38001f0d8c572d69a>
- Prerequisite page:
 - Prerequisites
 - Understand Python, Linear Algebra, Statistics
- Course Schedule
 - French section: <https://fracturedplane.notion.site/Cours-de-Science-des-Donn-es-IFT6758B-2026-3c1214857276818499f7e8eee49dc68a?pvs=74>
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Second Step: Piazza

- Discussion and Questions on Piazza: Email already sent.
- We will not use Studium!
- Make sure you are registered to Piazza! Otherwise fill [this form]
- Make sure to check Piazza frequently.
 - and check your @umontreal.ca email!

Third Step: Survey

- Fill the survey [link here](#)
- Super important!
 - This will help us focus the course to your background.
 - Organize project teams
- I will leave you more time for the break to fill the form!

Course Logistic: Gradescope

- Quizzes on Gradescope.
- Coding homework will be provided but not graded. However, it will appear on written exams.
- Quiz 0 next week! (to practice how to submit answers)

Course Logistics: Grading

- Final Project (3 milestones) (35%)
- Class participation (5%)
- Midterm + Final (written exams)
 - Midterm: Week after reading week (25%)
 - Final in December (30%)
- Programming Homeworks (Gradescope)– (5%)
 - Examples on exams

The Final Project

- Let's Break the ice!



The Final Project

- NHL data publicly available.
- Getting the data from an API.
- Some data visualization.
- Some basic ML stuffs.
- Challenge: Non-standard domain -> real world data
- Goal: Production like pipeline



Why this Course?

- Data Science is changing fast!
- This course is designed to catch everyone up to the most advanced methods
- This course is created to help people get a job! Pass interviews.
 - We get what we put into our studies

Thoughts ON Generative Text Models

- How many have used ChatGPT/Gemini/Claude/Qwen/etc?
- What do you use them for?
- Do they always work?
 - Examples of them failing...

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- How many have used ChatGPT/Gemini/Claude/Qwen/etc?
- What do you use them for?
- Do they always work?
 - Examples of them failing...
- LLMs change
 - Models get updated, and their performance changes
 - Restricted use at work. Can't use them for sensitive projects
 - Use will be restricted or \$\$\$
- Truth is that many companies want to use AI, but don't know how.
 - They need experts who understand how to integrate AI into their systems
RELIABLY.

LLMs are Tools, Experts can Do Better With Tools

Example: A paint brush

- In an inexperienced hand, a paintbrush just mixes every color into the same muddy, boring tone.



- In an expert's hand, the same tool produces masterpieces.



- Vincent van Gogh, *The Starry Night* (1889)



- Leonardo da Vinci, *Mona Lisa* (c. 1503)

AI is being used Everywhere

- LLMs produce more not better. Therefor, we need to be better.
- However, AI can be used properly.
- CERN is using to help process data, improve code, etc
- Canadian Gov is using for data processing.
- Many companies improving on these tools
 - All work better with domain knowledge.

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- Questions?

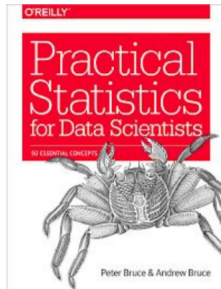
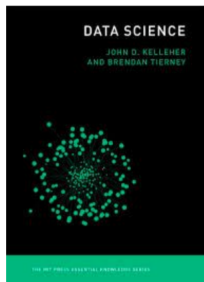
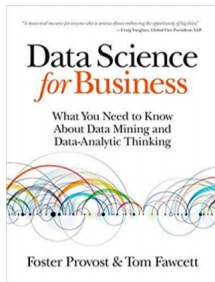
Poll: LLMs usage

Poll

- When is it good to use language models for our work?
 - (A) : When we're not sure what to do.
 - (B) : When we have some thoughts on how to complete the task, but not completely sure.
 - (C) : We're pretty sure how to complete the task. We just need something to help us auto-complete the process.
 - (D) : The tasks we've done many times before, but this is just a unique case, with a small difference in one of the prior times we've done the task.

Useful Resources

- Books: Statistical Learning, Python Handbook, Introduction to Data Science
- Online courses: freeCodeCamp , Harvard CS109
- Harvard's Data Science Review



Resources

- Some content from (<https://ucbrise.github.io/cs294-ai-sys-sp22/>,
- <https://fullstackdeeplearning.com/spring2021/>,
- <https://stanford-cs329s.github.io/>).

- Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*, 2022.
- Devvrit Khatri, Lovish Madaan, Rishabh Tiwari, Rachit Bansal, Venkata Sai Surya Subramanyam Duvvuri, Manzil Zaheer, Inderjit Dhillon, David Brandfonbrener, and Rishabh Agarwal. The art of scaling reinforcement learning compute for llms. In *International Conference on Learning Representations*, volume 2026, pages 72438–72467, 2026.
- Rui Li, Deji Fu, Chunyu Shi, Zhilan Huang, and Gang Lu. Efficient llms training and inference: An introduction. *IEEE Access*, 13:32944–32970, 2024.